



A behavioural comparison of route choice on metro networks: Time, transfers, crowding, topology and socio-demographics



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ARTICLE INFO

Article history:

Received 8 August 2013

Received in revised form 19 May 2014

Accepted 20 May 2014

Available online 12 June 2014

Keywords:

Decision making

Route choice

Metro networks

Network topology

ABSTRACT

Understanding travellers' behaviour is key element in transportation planning. This article presents a route choice model for metro networks that considers different time components as well as variables related to the transferring experience, train crowding, network topology and socio-demographic characteristics. The route choice model is applied to the London Underground and Santiago Metro networks, to make a comparison of the decision making process of the users on both cities. As all the variables are statistically significant, it is possible to affirm that public transport users take into account a wide variety of elements when choosing routes. While in London the travellers prefer to spend time walking, in Santiago is preferable to spend time waiting. Santiago Metro users are more willing to travel in crowded trains than London Underground users. Both user groups have a similar dispreference to transfers after controlling for the time spent on transfer, but different attitudes to ascending and descending transfers. Topological factors presented on a distorted Metro map are more important than actual topology to passengers' route choice decisions.

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1. Introduction

Understanding how public transport users make their travel decisions and being able to predict their behaviour is essential in transportation planning. The purpose of this study is to advance our understanding of the behaviour of public transport users when choosing a route in a Metro network and to quantify the impacts of the underlying explanatory variables that influence their decisions.

Route choice models have been explored and developed for private transport networks (Bovy and Stern, 1990; Ramming, 2001; Prato, 2009), but not much work has been done in public transport networks (Hunt, 1990; Bovy and Hoogendoorn-Lanser, 2005). The route choice variables normally included in traditional route choice models (either on private transport of public transport networks) limit to some basic service levels attributes of the alternative routes, such as travel time and fare (Ortúzar and Willumsen, 2011). However, other variables, related to both the level of service and the travellers' perceptions, influence the user's route choice process but are generally ignored in traditional modelling. This study presents a route choice analysis on Metro networks, incorporating variables related to the different times involved (travel, waiting and

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walking times), trains and stations usage, transfer environment, network topology and socio-demographic information from the travellers.

This study also conducts an empirical analysis to compare route choice decision making on the London Underground system and the Santiago Metro system, using the same modelling approach and specification. Even though behavioural comparisons can be made between studies found on the literature (mainly between results such as values of time and demand elasticities), these can generally be made based on models with different specification and context. One of the main objectives of this study is to develop a common framework to analyze and compare the preferences of travellers from both networks, and provide general transportation planning insights from the comparison.

The rest of the paper is organized as follows. In Section 2 we address the modelling approach and present the modelling variables considered. In Section 3 we present and discuss the route choice results for the London Underground and Santiago Metro networks. Finally, in Section 4 we present our main conclusions.

2. Route choice modelling

In this study we seek to identify and quantify the different aspects of travelling that are taken into account by public transport users from two cities (London and Santiago), particularly when choosing their travel routes. For this, it is essential to understand and model their decision making process, based on the different characteristics of the alternative routes.

2.1. Discrete choice models and correlation

We consider a random utility model, where it is assumed that each traveller q chooses a route i among a set $A(q)$ of available alternatives in order to obtain the maximum possible utility level U_{iq} . It is also assumed that the modeller, who is just an observer without perfect information regarding the decision making process, is only able to define a representative utility level V_{iq} . Thus, it is necessary to associate an error term ε_{iq} to each alternative (McFadden, 1974), typically as shown in (2.1).

$$U_{iq} = V_{iq} + \varepsilon_{iq} \quad (2.1)$$

The representative utility level V_{iq} is a function of different attributes X_{ikq} related to the routes and the travellers (e.g. travel times, transfer characteristics and travellers' perceptions). Generally, V_{iq} is assumed to be a linear function of the attributes, as shown in (2.2), where θ_{ik} are parameters to be estimated.

$$V_{iq} = \sum_k \theta_{ik} \cdot X_{ikq} \quad (2.2)$$

To characterize the individual decisions, binary variables d_{iq} that take values according to (2.3) are also needed. These binary variables correspond to the actual decisions made by the travellers.

$$d_{iq} = \begin{cases} 1 & \text{if } U_{iq} \geq U_{jq}, \quad \forall j \in A(q) \\ 0 & \text{in other case} \end{cases} \quad (2.3)$$

If the random terms ε_{iq} are assumed to be i.i.d. Gumbel, a Multinomial Logit (MNL) model is obtained, from which is possible to obtain an analytical expression for the choice probabilities P_{iq} , given by (2.4).

$$P_{iq} = \frac{\exp(V_{iq})}{\sum_{j \in A(q)} \exp(V_{jq})} \quad (2.4)$$

One of the limitations of the MNL model is that it does not consider correlation between alternatives. This may be particularly serious when modelling route choices, as strong correlation between the alternative routes may arise due to overlapping. To address this issue, different models have been proposed. We consider a C-Logit model (Cascetta et al., 1996), which includes a "commonality factor" CF_i in the MNL utility function to capture correlation between alternatives, as shown in (2.5). The inclusion of the commonality factor helps in correcting the models' fit and predictions by lowering the probabilities of choosing similar alternatives.

$$P_{iq} = \frac{\exp(V_{iq} + \beta \cdot CF_i)}{\sum_{j \in A(q)} \exp(V_{jq} + \beta \cdot CF_j)} \quad (2.5)$$

β is a negative parameter that captures the travellers perception towards correlated alternatives (e.g. if β equals zero, all routes are considered independent; if β is large in magnitude, independent routes will tend to be chosen over correlated routes). The commonality factor CF_i can be defined in many ways. The specification used in this study is shown in (2.6), where l_a is the length of link a , L_i is the length of route i , and δ_{aj} is equal to 1 if link a belongs to route j or 0 otherwise. The proposed specification resulted in a slightly better goodness-of-fit than others mentioned in Prato (2009).

$$CF_i = \ln \sum_{a \in i} \left(\frac{l_a}{L_i} \cdot \sum_{j \in A(q)} \delta_{aj} \right) \quad (2.6)$$

Having the attributes X_{ikq} and the choices d_{iq} , the parameters θ_{ik} and β can be estimated using maximum likelihood.

2.2. Metro networks for analysis

We conduct an empirical analysis to compare route choice decision making on the London Underground system (based on the work of [Guo, 2011](#)) and the Santiago Metro system (based on the work of [Raveau et al., 2011](#)). Both systems have high ridership (3 million daily trips in London Underground and 2.3 million daily trips in Santiago Metro) but differ in size (London Underground has 402 km of length, while Santiago Metro only 103 km) and complexity. Based on their characteristics, the results from the comparison could be generalized to public transport systems in general. Also, as both systems conduct similar route choice surveys for planning purposes, we can specify and estimate the same utility function on both networks to compare the behaviour characteristics of the respective travellers.

The London's trip database is based on data collected by Transport for London (TfL) from 1998 to 2005. During the period of analysis, the London Underground network had 11 lines, 255 stations and 72 transfer stations. The database consists of 17,073 individual trips along the day between 2127 OD pairs (3% of the total OD pairs of the network) with two or more alternative routes. The details of the database can be found in [Guo and Wilson \(2011\)](#).

The Santiago's trip database was obtained from an origin–destination survey conducted at Metro stations in October 2008. When the survey was conducted the network consisted of 5 Metro lines, 85 stations and 7 transfer stations. The database obtained from the survey consists of 28,961 individual trips along the day between the 1365 different OD pairs (19% of the total OD pairs of the network) which have two or more alternative routes to travel (i.e. those OD pairs where there is a route choice decision). A specific analysis of the peak-hours travels can be found in [Raveau et al. \(2011\)](#).

2.3. Topology and map effects

One key element to be analyzed is the influence of the network's topology on the route choice, as travellers might tend to prefer those alternatives that seem more direct between the origin and destination stations (aside from other variables such as travel time). To test this hypothesis we define an angular cost ([Raveau et al., 2011](#)) to measure how direct/indirect a given route is.

On most public transport systems in the world the network is presented to the users through display maps, where all the relevant information for travelling (different lines, stations, transfer points, etc.) is included. To enhance the understanding and make the maps easier to read, the public transport network tends to be distorted in the maps ([Ovenden, 2007](#)). As public transport maps usually do not include service information (such as travel time or crowding levels), their distortion might affect the decisions made by the travellers.

[Fig. 1](#) shows the true geographic (panel *a*) and display map (panel *b*) depictions of the London Underground and Santiago Metro networks. On both cases the map displayed in the stations and consulted (or at least seen) by users on a day-to-day basis contains serious distortions, both in the relative location of the stations and the distances between them. This way, users' angular and geometric perceptions may also diverge from the geographical reality, which may induce them to choose routes that are not the best for them.

Although the Santiago Metro network has grown considerably over the last decade, and will continue growing, it is still far away of being as complex and dense as the London Underground network. Also, the distortion is less severe (the correlation between the true and map distances in Santiago Metro is 94%, while in London Underground is only 22%). This way, the topology and map impacts on the travel decisions should be more significant in London.

2.4. Set of alternative routes

When dealing with probabilistic route choice models (such as a C-Logit model) it is fundamental to explicitly define an appropriate set $A(q)$ of available alternatives. This is not trivial, as there is no sure way of knowing which routes were considered by the travellers but were not chosen. For this, many approaches (both deterministic and stochastic) have been developed for constructing sets of alternative routes, trying to replicate the mental selection process undertaken by travellers (for a complete review on these approaches see [Prato, 2009](#)).

For the Santiago Metro study case the set of alternative routes for each OD pair was generated based on all the routes chosen by the travellers of that OD pair. This was possible due to the large ratio between trips and OD pairs (21.2 trips per OD pair, in average), so in every one of the 1365 considered OD pair there were at least two routes used. This way, no methodological approach for generating alternative routes was necessary, and the set obtained comes directly from the observed traveller's behaviour. This procedure is not common in the literature (in most surveys the number of trips per OD pair tends to be low), but it can be considered an ideal scenario, as no exogenous assumptions must be made regarding how travellers select alternative routes. Nevertheless, if choice set generation approaches were to be applied, the resulting choice set would be very similar to the one obtained by observing the travellers choices, as in this case the network is not particularly big or dense.

On the London Underground study case the ratio between trips and OD pairs is lower (7.7 trips per OD pair, in average) and in most OD pairs there is only one chosen route. Therefore, a methodological approach for obtaining a set of available alternatives is needed. A labelling approach ([Ben-Akiva et al., 1984](#)) was used to generate the set of available alternatives, by

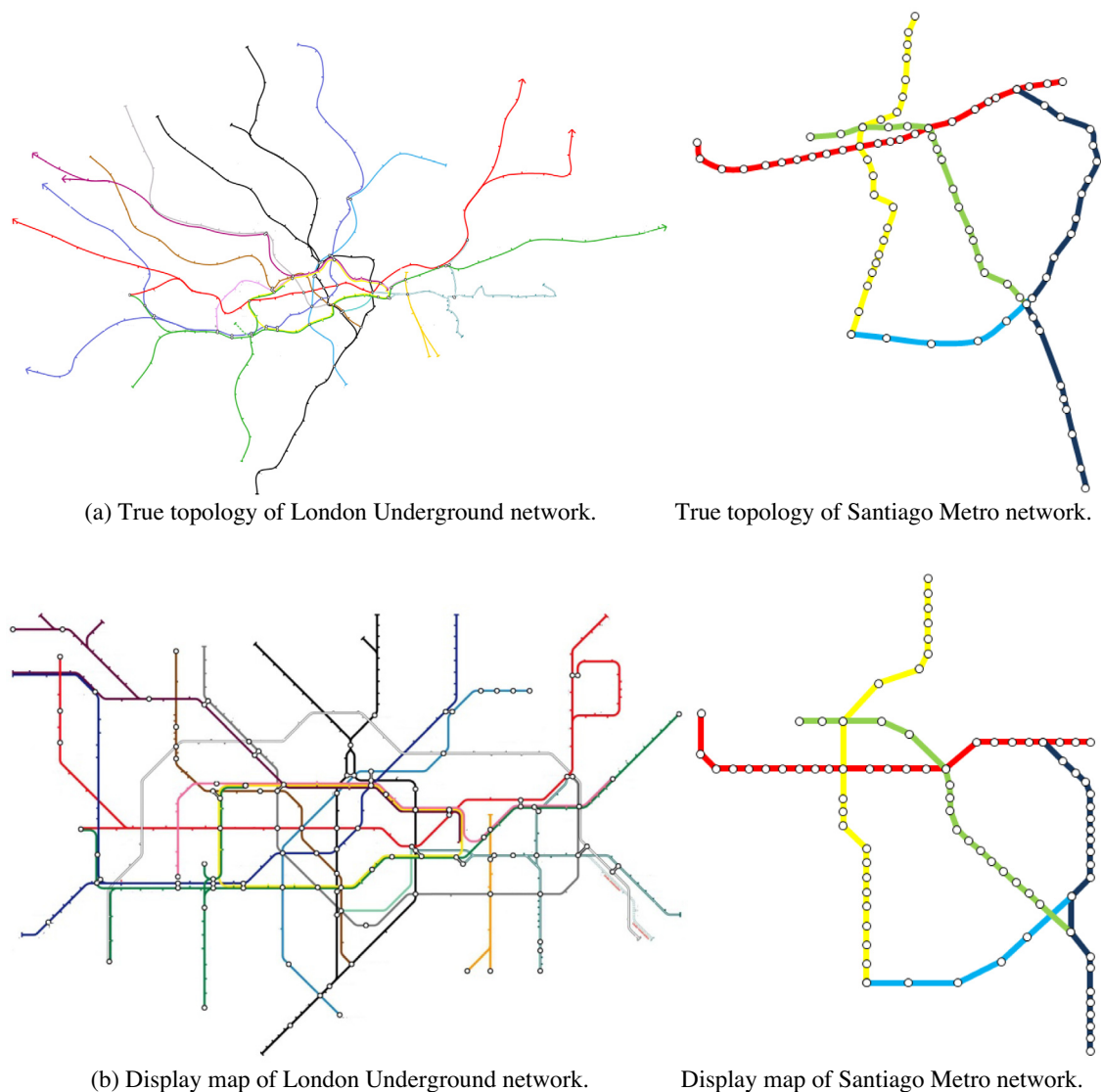


Fig. 1. London Underground and Santiago Metro networks topologies.

obtaining the shortest path for different definitions of “link costs”. In this case, different combinations of weights for the time components (travelling, waiting, walking and dwelling), transfer penalties and reliability measures were considered. Details about the application of the method and its results (such as efficiency, effectiveness and coverage) can be found in [Guo and Wilson \(2011\)](#). A minimum number of two and up to six different routes were obtained for each of the 2127 OD pairs. The main advantage of using a labelling approach lies on its behavioural assumptions, as different route selection strategies can be addressed (e.g. some travellers might consider the fastest route, while others might want a route that is good in terms of comfort) and every route is optimal in some sense.

The distribution of OD pairs and trips according to the number of available routes is presented in [Table 1](#). As the London Underground network is denser, there are more alternative routes for travelling than in the Santiago Metro network. It can be seen as well that in both cities the OD pairs distribution and the trips distribution is very similar: no particular OD structure (in terms of available alternatives) tends to concentrate more trips than another.

2.5. Model description

In the literature there are multiple route choice models (mainly MNL models or extensions of it) based on users' socio-economic characteristics and route attributes ([Ramming, 2001](#); [Prashker and Bekhor, 2004](#); [Liu et al., 2010](#)). However, the attributes included in these models are all tangible and quite limited to travel time components, fare and transfers. This

Table 1
Route availability statistics.

Available routes	Santiago Metro				London Underground			
	OD pairs		Trips		OD pairs		Trips	
2	1322	(96.9%)	26,950	(93.1%)	1161	(54.6%)	9589	(56.2%)
3	37	(2.7%)	1892	(6.5%)	694	(32.6%)	5559	(32.6%)
4	6	(0.4%)	119	(0.4%)	202	(9.5%)	1479	(8.7%)
5	0	(0.0%)	0	(0.0%)	58	(2.7%)	348	(2.0%)
6	0	(0.0%)	0	(0.0%)	12	(0.6%)	98	(0.6%)
Total	1365	(100.0%)	28,961	(100.0%)	2127	(100.0%)	17,073	(100.0%)

study presents a more complete route choice analysis on Metro networks, incorporating variables related to the different times involved (travel, waiting and walking times), trains and stations usage, transfer environment, network topology and socio-demographics. The additional variables, included to improve the ability to explain and predict route choice decisions, are discussed below. Some of these additional variables were considered separately by Guo (2011) and Raveau et al. (2011); in this study we consider their joint effect when applied to two different metro networks. Thus, the main contributions of this paper when compared with the work previously conducted by Guo (2011) and Raveau et al. (2011), aside from the behavioural comparison of both cities (something rarely done in travel demand literature), are the inclusion of socio-demographic variables and route correlation. It's important to take into account that no fare variable was included because all alternative routes between a given OD pair have the same monetary cost.

The most traditional and important variable used to explain route choice behaviour is the travel time. Users tend to look for the fastest way of getting from their origin to their destination, and the travel time is the main criterion to discard unattractive (i.e. slow) alternatives. We consider three different time components: the *in-vehicle time*, the *waiting time* at the origin station and all subsequent transfer stations, and the *walking time* when transferring (as the origins and destinations are fixed, we obviate access and egress times). These different time components are considered separately to address their different perception and importance in the travellers' decision making process.

Regarding the transferring experience, we firstly consider the total *number of transfers* of each alternative route; as the actual transferring time is captured by the walking and waiting time variables, this variable solely captures the displeasure of having to transfer. To further understand the transferring valuation, we differentiate between possible types of transfers. In terms of stations layout, the transfers can either be made between *ascending levels* (i.e. going up), *even levels* (usually walking across the platform) and *descending levels* (i.e. going down). In terms of stations infrastructure, the transfers can be *assisted* (made completely using escalator and/or lift), *semi-assisted* (made partially using escalator and/or lift and partially on foot), and *non-assisted* (made completely on foot).

To address the level of comfort and crowding experienced by the public transport users during their trip, the *mean occupancy* along the route was included in the models. This variable is defined as the distance-weighted ratio between passengers load and train capacity in peak hour (even though not all trips are made in peak hour, this specification presented the best goodness-of-fit). By definition the rate can vary between 0 (trains travelling empty along all arcs of the route) and 1 (trains travelling fully loaded along all arcs of the route).

Two additional variables related to train usage and extreme crowding levels were included in the model. We distinguish those transfer stations where there is the *possibility of getting a seat*, depending on the occupancy of the trains leaving those stations. On London Underground this happens when the occupancy is 20% or less, while on Santiago Metro this happens when the occupancy is 15% or less (these percentages represent the percentage of the capacity that corresponds to seats). On the other hand, we distinguish those transfers where there is the *possibility of not boarding* the first train, and having to wait for the next train. We observe that on London Underground this happens when the occupancy is 70% or more, while on Santiago Metro this happens when the occupancy is 85% or more. Based on these thresholds, it may seem that Santiago Metro's users are more willing to board crowded trains, instead of simply waiting for the next one.

To deal with the topology's effect on the route choices of the travellers, we include different topological factors. Following Guo (2011), we include the *distance* and *number of stations* between the origin and the destination along the different routes. Due to variable dwelling times, spacing between stations and distortion of the schematic maps (on which the topological variables are based), these variables are not highly correlated with the travel time components and can be included in the models without problems. We also include an *angular cost* to measure how direct a certain route is. The angular cost is defined accordingly with Raveau et al. (2011) as shown by (2.7), where s represents a leg of the route, d_s is the distance of the leg s , and θ_s is the angle formed between the destination station, the first station of leg s and the last station of leg s . Travellers will tend to choose the routes that appear to be more direct between their origin and destination.

$$\text{Angular cost} = \sum_s d_s \cdot \sin\left(\frac{\theta_s}{2}\right) \quad (2.7)$$

Also related to the topology's effect, two additional variables were considered. We say that a route *turns back* if it has a transfer station (or the destination, if it is the last leg) that is closer to the origin than the transfer station immediately before it. We say that a route *turns away* if it has a transfer station that is further from the destination than the transfer station

(or the origin, if it is the first leg) immediately before it. These variables are an application of Dial's definition of reasonable routes (Dial, 1971), and capture the travellers reluctance to choose those routes that appear to be unreasonable in terms of geometry.

Finally, the socio-demographic information included in both databases consists of the *purpose of the trip* (we have restrictive purposes like going to work and going to study, and non-restrictive purposes like going home and going shopping), the *gender* and *age* of the traveller, *fare type* (full fare or discount pass), and *time of the day* the trip is made. All these variables were also considered for formulating interactions with the rest of the explanatory variables previously described.

Information regarding the *frequency of the journey* (e.g. daily, weekly, monthly, first time ever) was available only for London and *income level* was available only for Santiago; these two variables were excluded from the modelling for comparison purposes. It is important to notice that the income level, a traditionally important variable in transport modelling, was not relevant anyway when modelling the decisions in Santiago Metro, as no monetary cost is included as an explanatory variable (therefore, there is no income effect in the route choices).

3. Results and analysis

Based on the data gathered on route choice in the London Underground and Santiago networks, C-Logit models were estimated to understand the travellers' decision making process. Other models that consider route correlation were also estimated (in particular, the Path Size Logit Model, the Cross Nested Logit, and the Paired Combinatorial Logit), but the C-Logit resulted on the best goodness-of-fit for both networks. When specifying the model, all topological variables were defined based on the display maps of the networks, rather than based on their true topology, as they resulted on better results (this issue will be further treated on Section 3.2). The estimated parameters, their *t*-values and goodness-of-fit indicators for the model are presented in Table 2. For the explanatory variables that interact with socio-demographic variables, the base parameter and marginal parameters are presented; this way, for example, the parameter of in-vehicle time for a trip on the morning peak period with a restrictive purpose in Santiago would be $-0.074 - 0.014 - 0.025 = -0.113$.

Table 2
Parameters estimates.

Attribute	London Underground		Santiago Metro	
	Parameter	t-Value	Parameter	t-Value
In-vehicle time	-0.121	-9.20	-0.074	-6.30
+Morning peak	-0.084	-5.10	-0.014	-2.52
+Afternoon peak	n.a. ^a	n.a.	-0.009	-2.62
+Restrictive Purpose	-0.042	-2.42	-0.025	-5.93
Waiting time	-0.269	-14.21	-0.083	-3.62
+Morning peak	-0.208	-5.94	-0.094	-2.60
Walking time	-0.299	-9.32	-0.210	-2.34
+Women	-0.048	-2.06	-0.074	-2.67
Number of transfers	-1.321	-4.14	-0.662	-4.19
Ascending transfers	-0.206	-2.53	-0.308	-2.60
Even transfers	0.613	3.82	n.a. ^b	n.a.
Descending transfers	0.000 ^c	n.a.	0.000 ^c	n.a.
Assisted transfers	0.000 ^c	n.a.	0.000 ^c	n.a.
Semi-assisted transfers	-0.271	-5.30	n.a. ^b	n.a.
Non-assisted transfers	-0.398	-6.33	-0.182	-5.11
Mean occupancy	-2.898	-3.25	-0.935	-5.10
Getting a seat	0.117	2.22	0.105	3.68
Not boarding	-0.502	-6.23	-0.358	-2.29
Angular cost	-0.088	-3.89	-0.029	-3.84
+Restrictive purpose	0.049	3.79	0.011	2.70
Map distance	-0.364	-5.43	-0.278	-4.83
Number of stations	-0.424	-5.07	-0.168	-3.62
Turning back	-0.650	-8.85	-0.142	-8.10
Turning away	-0.943	-7.77	-0.231	-8.87
Commonality factor	-0.396	-3.74	-0.541	-3.41
Sample size	17,073		28,961	
Log-likelihood	-6690		-12,881	
Corrected ρ^2	0.567		0.383	

^a This parameter is not statistically significant.

^b These kinds of transfers do not exist in Santiago Metro.

^c These parameters were fixed as base categories for the layout and infrastructure variables.

All the parameters presented in Table 2 have the expected signs and are statistically significant at 95% confidence level. For the station layout, the descending transfer parameter was fixed in zero as the base category; for the station infrastructure, the assisted transfer was defined as base category. It can be seen that an even transfer is preferred over changing levels; if a change in levels must be made, users prefer to descend rather than ascend, as ascending seems to be mentally associated with a greater effort. In terms of infrastructure, as the grade of assistance increases, the transfer experience improves. The parameter of the commonality factor, related to route correlation due to overlapping, has a similar statistical significance on both cities. This means that, at some degree, travellers in London and Santiago tend to perceive overlapping routes as non-independent alternatives.

For both cities, the socio-demographic are most relevant when interacted with the travel time components. There is also a significant interaction between the angular cost and the purpose of the trip (those travelling to work or study are more likely to be commuters, who are less influenced by the map). Other interactions, particularly with the variables related to transferring and crowding, are not statistically significant in the presence of time interactions. On both cities the socio-demographic variables of age and fare type (which are correlated, as fare discount are available for students and elders) did not provide any significant interaction with the explanatory variables, which is somehow surprising given the rich and broad trip sample used for both cities. The interaction of afternoon peak time and in-vehicle travel time was significant only for Santiago. It is worth noticing the surprising similarity of the explanatory and socio-demographic variables that are relevant when understanding the route choices in London and Santiago.

3.1. Perceptions and valuations

The parameters obtained for the London Underground and Santiago Metro are not directly comparable between each other, as the MNL-based models (such as the C-Logit) have different scales from their Gumbel errors variances (Train, 2009), but marginal rates of substitution can be derived from these parameters and compared for both networks. These marginal rates are presented in Table 3. As the interaction between socio-demographic variables and explanatory variables provide multiple in-vehicle time parameters (4 in London and 6 in Santiago), we are only presenting and analysing the results for morning peak trips with restrictive purpose; the results for all other trips can be obtained analogously. For this direct comparison, the angular cost and the map distance are excluded as they do not have a measurement scale.

On both cities the waiting and walking times are more valuable than the in-vehicle travel time, but while in London the value of the waiting time is greater than the value of the walking time, in Santiago the relationship is the opposite. When choosing transferring alternatives, Santiago Metro users will be more willing to wait and London Underground users will be more willing to walk, in part because Santiago Metro has fewer transfer options and their environment is generally simpler, while London Underground presents many transfer alternatives and the connection between platforms could be longer and more complex due to the incremental addition of lines over the time.

Even though in both cities the valuation of a base transfer (this is, an assisted descending transfer where there is no possibility of either getting a seat or not being able to board) might seem very similar, it's necessary to take into account the marginal valuations of the layout and infrastructure variables. As in London Underground the most used parts of the network are underground, a descending change of levels will tend to have associated a corresponding ascending change of levels; this way the valuation of ascending transfers is lower in London Underground than in Santiago Metro. For example, while London Underground is completely underground in the city centre of London, in the city centre of Santiago almost 30% of the Santiago Metro network is overground.

Table 3
Marginal rates of substitution – morning peak & restrictive purpose.

Attribute	London Underground	Santiago Metro
1 min of waiting time	1.93 min in-vehicle	1.57 min in-vehicle
1 min of men walking time	1.21 min in-vehicle	1.86 min in-vehicle
1 min of women walking time	1.40 min in-vehicle	2.51 min in-vehicle
1 base transfer ^b	5.35 min in-vehicle	5.86 min in-vehicle
1 ascending transfer	0.83 min in-vehicle	2.73 min in-vehicle
1 even transfer	2.48 min in-vehicle ^a	n.a. ^c
1 non-assisted transfer	1.61 min in-vehicle	1.61 min in-vehicle
1 semi-assisted transfer	1.10 min in-vehicle	n.a. ^c
1% of occupancy	0.12 min in-vehicle	0.08 min in-vehicle
Possibility of seating	0.47 min in-vehicle ^a	0.93 min in-vehicle ^a
Possibility of not boarding	2.03 min in-vehicle	3.17 min in-vehicle
1 station along the way	1.72 min in-vehicle	1.49 min in-vehicle
Turning back	2.63 min in-vehicle	1.26 min in-vehicle
Turning away	3.82 min in-vehicle	2.04 min in-vehicle

^a Absolute value, these variables represent a gain in utility.

^b Based on the model specification, this base transfer corresponds to an assisted descending transfer where there is no possibility of either getting a seat or not being able to board.

^c These kinds of transfers do not exist in Santiago Metro.

Table 4

Transferring valuations – morning peak & restrictive purpose.

Characteristics		Getting a seat (min)	Intermediate (min)	Not boarding (min)
<i>London Underground</i>				
Ascending	Assisted	5.71	6.18	8.21
Ascending	Semi-assisted	6.84	7.28	9.31
Ascending	Non-assisted	7.32	7.79	9.83
Even		2.39	2.87	4.90
Descending	Assisted	4.87	5.35	7.38
Descending	Semi-assisted	5.97	6.45	8.48
Descending	Non-assisted	6.49	6.96	8.99
<i>Santiago Metro</i>				
Ascending	Assisted	7.66	8.58	11.75
Ascending	Non-assisted	9.27	10.19	13.36
Descending	Assisted	4.93	5.86	9.03
Descending	Non-assisted	6.54	7.47	10.64

To further analyze the users' valuation of transfer experiences, Table 4 presents the marginal rates of substitution for all possible transfer types (once again for morning peak trips with restrictive purpose). There is a high variability in London transfer's valuations (between 2.39 and 9.83 min of in-vehicle travel time) and Santiago transfer's valuations (between 4.93 and 13.36 min of in-vehicle travel time). This way, the necessity of distinguishing by station layout, infrastructure and occupancy is clear; otherwise the heterogeneity in preferences would not be captured. Weighting each valuation for the number of transfer of each type made on a regular day, the average transfer penalty in London Underground is 7.0 min and in Santiago Metro is 10.2 min. It seems that, in general, London Underground users are more willing to transfer than Santiago Metro users. As the London Underground network is bigger and denser, there are more transfer possibilities and Londoners are more used to transferring: 69% of the trips in London Underground involve at least one transfer, while in Santiago Metro is the 47% of the trips.

Also from Table 3 it can be seen that occupancy is more valued in London, as the London Underground users are less willing to travel in crowded trains than Santiago Metro users. This also explains why Londoners are more sensitive towards not boarding a train due to crowding, consistent with the lower threshold obtained when defining the possibility of not boarding. As an example of this, TfL defines a "crowded train" as 4 passengers per m², while Santiago Metro defines a "crowded train" as 6 passengers per m². On the other hand, extreme occupancies (getting a seat or not being able to board) are more valued in Santiago, probably because in London the occupancies are lower.

Avoiding one station along the way (this is, taking a route with one station less between the origin and the destination) is valued similarly on both cities. To further understand this marginal rate, the mean travel time between two consecutive stations (considering dwelling time) in London Underground and Santiago Metro are 1.8 min and 1.6 min, respectively. The similarity between these values and the marginal rates obtained indicates that the public transport users have a good notion of the operation times of the networks.

Londoners are much more sensitive to taking routes that seem unreasonable in terms of their geometry, either because they turn back to the origin or turn away from the destination. This could be due to the high distortion of the London Underground map. Another possible explanation is that the London Underground network generates more unreasonable alternative routes due to its complexity and density, which in reality are barely considered by the users.

3.2. Specification of topological variables

As mentioned before, the topological variables (in particular, the angular cost, the distance, and the impressions of turning back and turning away) can be specified either using the true topology of the networks or their display maps. Addi-

Table 5

Topological variables specification – morning peak & restrictive purpose.

Attribute	London Underground		Santiago Metro	
	True <i>t</i> -values	Map <i>t</i> -value	True <i>t</i> -values	Map <i>t</i> -value
Angular cost	−1.62	−3.89	−2.12	−3.84
+Restrictive purpose	1.34	3.79	2.03	2.70
Distance	−3.64	−5.43	−2.81	−4.83
Turning back	−4.02	−5.07	−3.97	−3.62
Turning away	−4.30	−8.85	−4.12	−8.10
Sample size	17,073		28,961	
Log-likelihood	−6716	−6690	−12,957	−12,881
Corrected ρ^2	0.565	0.567	0.379	0.383

tionally to the model specified using the display maps (whose complete results were presented in Table 2), the alternative model specified using the true topologies was also estimated. Table 5 summarizes some of the main differences between both models, by presenting the *t*-values of the topological variables (all other control variables are also included but we focus on these variables, as these are the ones that change in specification) and goodness-of-fit indicators. It's important to take into account that in the new model, specified with the true topologies, all parameters have the correct sign and that the parameters of the rest of the variables (those omitted in Table 5) are similar to the parameters obtained for the model specified using the display map.

It can be seen from Table 5 that all the topological variables have less statistically significant parameters when the true topologies are used to specify them. In particular, the angular cost turns out to be not statistically significant at 90% confidence level for the London Underground. Better goodness-of-fit is obtained when the topological variables are specified using the display maps information. This reassures the idea that public transport users take into account display maps (with which are more familiarized) rather than the actual topology of the networks. This strongly indicates the relevance of the way the information is provided to users, as any map distortion could induce them to make inadequate decisions.

3.3. Model specification and information omission

To analyze the importance of a correct and complete model specification, five additional models with different amount of variables were estimated: one without the occupancy-related variables, one without the transfer-related variables, one without the topology-related variables, one without the socio-demographic variables, and without the four types of variables (this is, considering only the time-related variables). Even though all models have correct parameters' signs, some serious problems in terms of bias can arise. Table 6 presents the marginal rates of substitution between time components and the global goodness-of fit indicator for all models.

It can be seen that, when omitting information by excluding explanatory variables, there is bias in the marginal rates of substitution obtained. The incomplete models tend to overestimate the valuations of waiting and walking times; this is expected when, for example, transfer variables are omitted, as these two time components (directly related with the transfer experience) must capture the disutility of transferring. When the topological variables are excluded from the London model, underestimations of the marginal rates of substitution are obtained, while in Santiago the marginal rates are overestimated. One possible explanation might be the topological variables tend to guide Londoners to choose transfer alternatives that involve longer walking and waiting time, so when these variables are excluded from model, it would appear that Londoners have a lower valuation on transfer walking and waiting. The opposite happens in Santiago, where the topological variables tend to guide passengers to choose transfer stations with shorter walking and waiting time.

In terms of goodness-of-fit, the worst results (aside from when all four subsets of variables are excluded) are obtained when the five topological variables are omitted from the model, rather than when the five transfer variables or the three

Table 6
Explanatory variables specification – morning peak & restrictive purpose.

Model	Time	London Underground	Santiago Metro
Complete model	Waiting	1.93 min in-vehicle	1.57 min in-vehicle
	Men's walking	1.21 min in-vehicle	1.86 min in-vehicle
	Women's walking	1.40 min in-vehicle	2.51 min in-vehicle
	Corrected ρ^2	0.567	0.383
Model without occupancy variables	Waiting	1.80 min in-vehicle	2.38 min in-vehicle
	Men's walking	1.29 min in-vehicle	1.90 min in-vehicle
	Women's walking	1.43 min in-vehicle	2.68 min in-vehicle
	Corrected ρ^2	0.558	0.373
Model without transfer variables	Waiting	2.57 min in-vehicle	3.82 min in-vehicle
	Men's walking	1.98 min in-vehicle	2.50 min in-vehicle
	Women's walking	2.31 min in-vehicle	3.11 min in-vehicle
	Corrected ρ^2	0.535	0.369
Model without topological variables	Waiting	0.61 min in-vehicle	2.30 min in-vehicle
	Men's walking	0.46 min in-vehicle	2.53 min in-vehicle
	Women's walking	0.53 min in-vehicle	3.77 min in-vehicle
	Corrected ρ^2	0.520	0.357
Model without socio-demographic variables	Waiting	1.66 min in-vehicle	1.46 min in-vehicle
	Men's walking	1.17 min in-vehicle	1.62 min in-vehicle
	Women's walking	1.17 min in-vehicle	1.62 min in-vehicle
	Corrected ρ^2	0.566	0.382
Model without occupancy, transfer, topological and socio-demographic variables	Waiting	0.80 min in-vehicle	4.48 min in-vehicle
	Men's walking	0.73 min in-vehicle	2.64 min in-vehicle
	Women's walking	0.73 min in-vehicle	2.64 min in-vehicle
	Corrected ρ^2	0.483	0.334

occupancy variables are omitted. This happens in both cities, and reassures the importance of the topology in the travellers' decision making process despite the very different network structure and complexity in the two systems.

3.4. Implications on transportation planning

Methodological comparisons between travel behaviour in different cities and contexts using the same approach are rarely done, mainly because limitations in terms of data access. Comparisons are generally between models with different specifications, different data collection methods, or different choice contexts. This way, the comparison of the route choice preferences in the London Underground and Santiago Metro networks can provide some findings about route choice preferences that might be extendable to any metro network, and also provide some insights for transportation planning.

One of the main findings of the study is the (somehow surprising) similitude of the results obtained for both cities, particularly in terms of which are the relevant variables when explaining the decisions of the travellers. Users of both systems consider different time components, transfer-related characteristics, crowding variables, and topological variables (which come from the respective schematic maps). Even the socio-demographic variables that interact (and do not interact) with the explanatory variables were found to be almost the same.

Having found so similar results in so different networks, we can argue that the explanatory variables included in this study can provide satisfactory results when analyzing and modelling route choice behaviour in any metro network, and can be a good starting point when dealing with route choice in multi-modal transit networks.

It is important to notice that, although the relevant variables were found to be very similar in both networks, the specific preferences and valuations were generally very different in both cities. This way, none of the particular results or valuations should be transferred and used in other context; with the appropriate data, the specification proposed could be used to obtain specific attribute valuations.

Finally, from a social planning point of view, it has been shown and confirmed that travellers take into account a wide variety of attributes when choosing their routes, and that their preferences can vary depending on their gender, the time of the day or the purpose of the trip. Travellers do not only care about travel times and number of transfers, but also care about crowding, transfer characteristics and topological factors. These results should be considered by the authorities and the planners, as many of the factors included in this study are not generally included in traditional route choice models. Particularly interesting is the impact of the schematic maps on the travellers' decisions, especially of those who are not familiar to the network (i.e. are not commuters).

4. Conclusions

Route choice modelling variables are traditionally limited to some tangible factors such as time and fare that, although relevant, fail to accommodate different aspects of traveller's behaviour. In this study we specify and estimate route choice models for metro networks that consider four different kinds of variables: travel time components, transfer experience characteristics, occupancy and comfort indicators, and network topology variables. All these variables result significant understanding for travellers' behaviour. This reassures the idea that public transport users take into account a wide variety of attributes when choosing routes.

An incomplete model specification can result on biased results, especially attributes valuations, which could have a great repercussion when valuating transportation projects (such as network expansions or operational changes). Also related to the specification, the topological variables have a greater explanatory power when specified based on the networks' display maps rather than their true topology. Although distortions in these display maps can induce route choice decisions that reduce utility for some users, they could also be exploited by planners to induce optimal route choices to make better use of public transport system capacity (Jankowski et al., 2001). This is particularly relevant in London Underground, where the map distortion is extremely big (probably bigger than any other public transportation system in the world). Schematic maps can also be complemented with information regarding level of service, to enable a more informed decision by the travellers (Hochmair, 2009).

As the route choice model was estimated for the London Underground and Santiago Metro networks, a comparison between traveller's behaviour in both cities was possible. London Underground users seem to be more willing to transfer than Santiago Metro users, as they are more used to required transfers when travelling. When transferring in London, users will tend to prefer to spend more walking than waiting, while in Santiago users will tend to prefer to waiting time over walking time. In terms of occupancy and comfort, Londoners are less willing to travel in crowded trains and will tend to stop boarding when the trains reach a lower threshold than in Santiago; on the other hand, they care less about getting a seat than Santiago Metro users.

Acknowledgments

We wish to thank the Chilean Fund for Promotion of Scientific and Technological Development, FONDEF (Project D10I1049), the Chilean Fund for Scientific and Technological Research, FONDECYT (Project 1110720), the VREF Centre of Excellence on BRT, the Centre for Sustainable Urban Development (CEDEUS), Transport for London and Metro de Santiago.

References

- Ben-Akiva, M.E., Bergman, M.J., Daly, A.J., Ramaswamy, R., 1984. Modeling inter-urban route choice behaviour. In: Volmuller, J., Hamerslag, R. (Eds.), *Proceedings of the 9th International Symposium on Transportation and Traffic Theory*. VNU Science Press, Utrecht, The Netherlands, pp. 299–330.
- Bovy, P.H.L., Hoogendoorn-Lanser, S., 2005. Modelling route choice behaviour in multi-modal transport networks. *Transportation* 32, 341–368.
- Bovy, P.H.L., Stern, E., 1990. *Route Choice. Wayfinding in Transport Networks*. Kluwer Academic Publishers, Norwell, MA.
- Cascetta, E., Nuzzolo, A., Russo, F., Vitetta, A., 1996. A modified Logit route choice model overcoming path overlapping problems: specification and some calibration results for interurban networks. In: Lesort, J.B. (Ed.), *Proceedings of the 13th International Symposium on Transportation and Traffic Theory*. Pergamon, Lyon, France, pp. 697–711.
- Dial, R.B., 1971. A probabilistic multipath traffic assignment model which obviates path enumeration. *Transp. Res.* 5, 83–113.
- Guo, Z., 2011. Mind the map! The impact of transit maps on path choice in public transit. *Transp. Res.* 45A, 625–639.
- Guo, Z., Wilson, N.H.M., 2011. Assessing the cost of transfer inconvenience in public transport systems: a case study of the London Underground. *Transp. Res.* 45A, 91–104.
- Hochmair, H., 2009. The influence of map design on route choice from public transportation maps in urban areas. *Cartogr. J.* 46, 242–256.
- Hunt, J.D., 1990. A logit model of public transport route choice. *ITE J.* 60, 26–30.
- Jankowski, P., Andrienko, N., Andrienko, G., 2001. Map-centred exploratory approach to multiple criteria spatial decision making. *Int. J. Geogr. Inf. Sci.* 15, 101–127.
- Liu, Y., Bunker, J., Ferreira, L., 2010. Transit users' route-choice modelling in transit assignment: a review. *Transp. Rev.* 30, 753–769.
- McFadden, D., 1974. Conditional Logit analysis of qualitative choice behavior. In: Zarembka, P. (Ed.), *Frontiers of Econometrics*. Academic Press, New York, USA, pp. 105–142.
- Ortúzar, J.de D., Willumsen, L., 2011. *Modelling Transport*. John Wiley and Sons, Chichester.
- Ovenden, M., 2007. *Transit Maps of the World*. Penguin Group, New York.
- Prato, C.G., 2009. Route choice modeling: past, present and future research directions. *J. Choice Model.* 2, 65–100.
- Prashker, J.N., Bekhor, S., 2004. Route choice models used in the stochastic user equilibrium problem: a review. *Transp. Rev.* 24, 437–463.
- Ramming, M.S., 2001. *Network Knowledge and Route Choice*. Unpublished Ph.D. Thesis, Massachusetts Institute of Technology, Cambridge, MA.
- Raveau, S., Muñoz, J.C., de Grange, L., 2011. A topological route choice model for Metro. *Transp. Res.* 45A, 138–147.
- Train, K., 2009. *Discrete Choice Methods with Simulation*. Cambridge University Press, Cambridge.